

### [Assignment-3]

Name : Imtiaz Hossain

ID : 23101137

Section: 30

Course : CSE 422  
Code

Faculty : [MZW]  
Initial

# Ans to the question no 1

a) Joint probability distribution table:

For the events A, B, and C

| A     | B     | C     | Probability |
|-------|-------|-------|-------------|
| True  | True  | True  | 0.2         |
| True  | True  | False | 0.1         |
| True  | False | True  | 0.15        |
| True  | False | False | 0.05        |
| False | True  | True  | 0.05        |
| False | True  | False | 0.05        |
| False | False | True  | 0.35        |
| False | False | False | 0.05        |

|          | C    |          | $\neg C$ |          |
|----------|------|----------|----------|----------|
|          | B    | $\neg B$ | B        | $\neg B$ |
| A        | 0.2  | 0.15     | 0.1      | 0.05     |
| $\neg A$ | 0.05 | 0.35     | 0.05     | 0.05     |

$$\text{Sum} = 0.2 + 0.15 + 0.05 + 0.05 + 0.35 + 0.05 = 0.85$$

∴ Missing probabilities for 2 and 8 must sum to

$$(1 - 0.85) = 0.15$$

$$P(A) = 0.5$$

$$P(A) = P(A, B, C) + P(A, B, \neg C) + P(A, \neg B, C) + P(A, \neg B, \neg C)$$

$$= 0.2 + x + 0.15 + 0.05 \quad [\text{Let, } P(A, B, \neg C) = x]$$

$$= 0.4 + x$$

$$\therefore 0.4 + x = 0.5$$

$$\Rightarrow x = 0.1$$

$$\therefore P(A, \neg B, C) = 0.1$$



Now, we have  $P(\neg C) = 0.25$

$$\begin{aligned} P(\neg C) &= P(A, B, \neg C) + P(\neg A, B, \neg C) + P(A, \neg B, \neg C) + P(\neg A, \neg B, \neg C) \\ &= 0.1 + 0.05 + 0.05 + y \quad [\text{Let, } P(\neg A, \neg B, \neg C) = y] \\ &= 0.2 + y \end{aligned}$$

$$\therefore 0.2 + y = 0.25$$

$$\therefore y = 0.05$$

$$\therefore P(\neg A, \neg B, \neg C) = 0.05$$

b)

$$\begin{aligned} P(\neg B) &= P(A, \neg B, C) + P(A, \neg B, \neg C) + P(\neg A, \neg B, C) \\ &\quad + P(\neg A, \neg B, \neg C) \\ &= 0.15 + 0.05 + 0.35 + 0.05 \\ &= 0.60 \end{aligned}$$

c)

$$P(A|B, C) = \frac{P(A, B, C)}{P(B, C)} = \frac{0.15}{0.15 + 0.35} = \frac{0.15}{0.50} = 0.3$$

$$P(\neg A|B, \neg C) = \frac{P(\neg A, B, \neg C)}{P(B, \neg C)} = \frac{0.05}{0.10 + 0.05} = \frac{0.05}{0.15} = \frac{1}{3} \approx 0.33$$

d)

$$P(A, B, C) = 0.2$$

$$\begin{aligned} P(A) \times P(B) \times P(C) &= 0.50 \times (1 - 0.60) \times (1 - 0.25) \\ &= 0.50 \times 0.40 \times 0.75 \\ &= 0.15 \end{aligned}$$

$$\therefore P(A, B, C) \neq P(A) \times P(B) \times P(C)$$

So, A, B, C are not independent

Ans to the question no 2

$$a) P(\text{Advanced}) = \frac{4}{15} \approx 0.2667$$

$$b) P(\text{None} \cap \text{Beginner}) = \frac{4}{15} \approx 0.2667$$

$$c) P(\text{Beginner} | \text{None}) = \frac{4}{5} = 0.8$$

d) Prior Experience = Some

Goal = professional

Availability = >10 hours

Beginner:

$$\begin{aligned} & P(\text{Beginner}) \times P(\text{Some} | \text{Beginner}) \times \\ & P(\text{Professional} | \text{Beginner}) \times \\ & P(>10 \text{ hours} | \text{Beginner}) \\ &= \frac{5}{15} \times \frac{1}{5} \times \frac{1}{5} \times \frac{1}{5} = \frac{1}{375} \approx 0.002667 \end{aligned}$$

Intermediate:

$$\begin{aligned} & P(\text{Intermediate}) \times P(\text{Some} | \text{Intermediate}) \times \\ & P(\text{Professional} | \text{Intermediate}) \times \\ & P(>10 \text{ hours} | \text{Intermediate}) \\ &= \frac{6}{15} \times \frac{3}{6} \times \frac{3}{6} \times \frac{3}{6} = \frac{1}{20} = 0.05 \end{aligned}$$



Advanced:

$$\begin{aligned} & P(\text{Advanced}) \times P(\text{some} | \text{Advanced}) \times \\ & P(\text{professional} | \text{Advanced}) \times \\ & P(>10 \text{ hours} | \text{Advanced}) \\ &= \frac{1}{15} \times \frac{1}{4} \times \frac{3}{4} \times \frac{2}{4} = \frac{1}{40} = 0.025 \end{aligned}$$

The highest numerator is for intermediate (0.05), so the suggested course difficulty is Intermediate.

e) The denominator in Bayes' Theorem, is the same for all classes when comparing posterior probabilities. Since it acts as a normalization constant, it does not affect the relative comparison of probabilities across classes. Therefore omitting the denominator and comparing only the numerators is sufficient for decision-making.

9) If a particular feature value never appears in the given class in the training data, the estimated conditional probability for that feature given the class is zero. Because Naive Bayes multiplies conditional probabilities, a single zero will make the entire product  $P(\text{features} | c)$  zero, forcing the posterior to zero even if other evidence strongly favors the class.

Example: Suppose no training example of class = Beginner. ever had prior = same.

Then,  $P(\text{prior} = \text{same} | \text{Beginner}) = 0$ . For

for any test instance with prior = same, the entire posterior score for Beginner becomes

zero.  $P(\text{Beginner}) = \frac{2}{4} = 0.5$

$$P(\text{professional} | \text{Beginner}) = \frac{1}{2} = 0.5$$

$$P(>10 \text{ hours} | \text{Beginner}) = \frac{1}{2} = 0.5$$

$$\text{Score}_B = 0.5 \times 0 \times \frac{1}{2} \times \frac{1}{2} = 0$$